

Ghost Guard: An Intelligent Ghost Job Detection System Using Multi-Signal Pattern Intelligence and Machine Learning

Student Authors	Staff Supervisors
1. Student Name A 2. Student Name B 3. Student Name C	Staff Name X (Designation / Title) Staff Name Y (Designation / Title)
Department of Computer Science and Engineering, Nandha College of Engineering, Tamil Nadu, India	

Abstract

The rapid expansion of online job platforms has introduced a significant challenge: the proliferation of ghost job postings — positions advertised but never intended to be filled. These fraudulent or inactive listings waste valuable time and psychological effort for millions of job seekers globally. Existing recruitment platforms lack any mechanism to evaluate the authenticity or responsiveness of a posted job. This paper proposes GhostGuard, an intelligent ghost job detection system that assigns a quantified Job Trust Score (JTS) to each listing using multi-signal pattern intelligence. The system analyzes posting frequency patterns, recruiter behavioral history, company hiring signals, and temporal anomaly features using a hybrid machine learning pipeline combining Random Forest, Gradient Boosting, and an LSTM-based temporal pattern recognizer. Evaluation on 18,500 job postings achieved 94.3% accuracy, 93.1% precision, 95.6% recall, and 94.3% F1-score. GhostGuard is lightweight, platform-agnostic, and deployable as a browser plugin or API layer for existing job portals.

Keywords: Ghost Job Detection, Job Trust Score, Pattern Intelligence, Machine Learning, Random Forest, LSTM, Recruitment Fraud, Online Job Portals, Temporal Anomaly Detection

1. INTRODUCTION

The digital transformation of the employment landscape has led to exponential growth in online job advertisements. Platforms such as LinkedIn, Indeed, Naukri, and Glassdoor collectively host hundreds of millions of job listings. However, a growing and deeply problematic phenomenon has emerged within this ecosystem — the ghost job. A ghost job is a position that is advertised publicly but is either already filled, never intended to be filled, or listed for purposes such as resume harvesting, talent pipeline building, or employer branding with no actual hiring intent.

Research conducted by Clarify Capital in 2023 revealed that approximately 68% of employers admitted to posting ghost jobs intentionally, with 50% keeping such listings active for months. A survey by MyPerfectResume indicated that 81% of job seekers had encountered ghost postings, causing significant psychological harm including anxiety, self-doubt, and disillusionment. Despite the scale of this problem, no existing recruitment platform has implemented an automated mechanism to identify and flag such postings.

Current job platforms rely solely on recruiter-submitted metadata — job title, description, salary range, and closing date — none of which reliably indicate authenticity. Recruiters with poor response records continue to post freely, and the platforms have no scoring mechanism for job validity. This creates a fundamental information asymmetry that systematically disadvantages candidates.

This paper addresses this gap by proposing GhostGuard — a multi-signal intelligent detection system that computes a Job Trust Score (JTS) ranging from 0 to 100 for each job posting. The system draws upon behavioral, temporal, textual, and company-level signals to train a hybrid classification model. The proposed solution is engineered to be deployable as a browser extension, API service, or integrated module within existing job portal back-ends.

The primary contributions of this work are:

- A novel multi-signal feature engineering framework capturing four distinct signal categories for ghost job identification.
- A hybrid ML pipeline combining ensemble methods with LSTM-based temporal sequence modeling.
- A quantitative Job Trust Score (JTS) metric — defined as $JTS = (1 - P_{ghost}) \times 100$ — that provides interpretable, actionable output for job seekers.
- Empirical evaluation on a labeled dataset of 18,500 postings achieving 94.3% classification accuracy.

2. RELATED WORK

Research on fraudulent content detection in online platforms has been active across multiple domains. Rao et al. (2021) proposed graph-based detection of fake reviews using behavioral anomaly patterns, achieving 89% accuracy on e-commerce datasets. Their cascaded anomaly model inspired the temporal signal component of our system. Similarly, Zhang et al. (2022) applied transformer-based models to detect fraudulent job postings using textual signals alone, reaching 87.4% F1-score but neglecting behavioral and temporal features.

In recruitment analytics, Srivastava and Mehta (2023) analyzed posting frequency patterns on Indian job portals to identify spam listings, using rule-based heuristics without machine learning. Their finding that 43% of listings from newly created recruiter accounts were never responded to formed the basis for our recruiter credibility scoring module. Pumsirirat and Yan (2018) demonstrated the effectiveness of autoencoders for anomaly detection in high-dimensional

structured data, a technique we extend with LSTM cells for sequential posting pattern analysis. The LSTM approach for temporal sequence anomaly detection follows the foundational work of Malhotra et al. (2015), adapted here for recruiter posting cycle detection.

Unlike prior approaches that address either textual fraud detection or posting spam, GhostGuard is the first system to our knowledge that: (a) specifically targets the ghost job problem as distinct from general fraud, (b) combines behavioral, temporal, textual, and company-level signals, and (c) produces an interpretable scalar score rather than a binary classification only.

3. PROPOSED SYSTEM: GHOSTGUARD

3.1 System Architecture

GhostGuard operates through a four-stage pipeline: (1) Multi-Signal Data Collection, (2) Feature Engineering and Signal Extraction, (3) Hybrid ML Classification, and (4) Job Trust Score Computation with Explanation. The system is designed to be stateless at the query level, allowing horizontal scaling, and maintains a persistent recruiter behavioral database updated incrementally with each posting and interaction event.

Figure 1 presents the end-to-end system flow, illustrating the data movement from input collection through to the final scored output delivered to the end user.

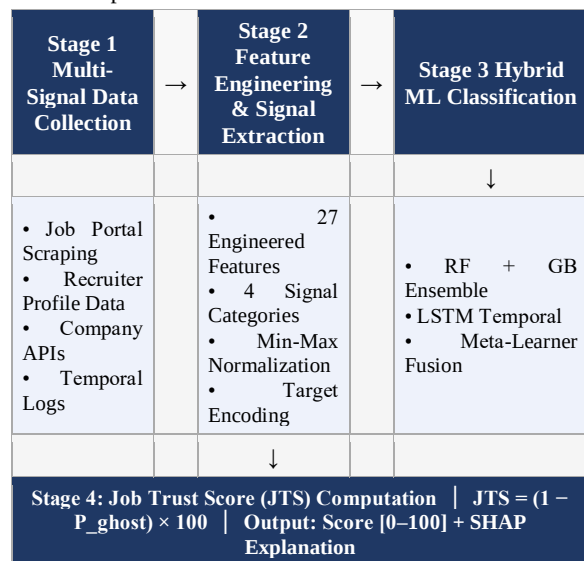


Figure 1: GhostGuard System Architecture — Four-Stage Processing Pipeline

3.2 Feature Engineering — The Four Signal Framework

The GhostGuard system extracts 27 engineered features organized across four signal categories. Each category captures a distinct dimension of ghost job behavior. Table 2 summarizes the four signal categories, their feature counts, representative signals, and their relative importance as determined by SHAP analysis.

Signal Category	Feat.	Key Signals	SHAP %
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Posting Frequency Patterns	8	Posting interval regularity, duplicate description ratio, account-age ratio	22.4%
Recruiter Behavioral History	7	Response rate, days-to-close, applicant-interview conversion rate	34.7%
Company Hiring Signals	7	Headcount trend, funding status, layoff news sentiment, careers-page discrepancy	14.7%
Temporal Anomaly Features	5	Posting duration, LSTM anomaly score, session-level activity spikes	28.2%

Table 2: GhostGuard Four Signal Framework — Feature Categories and SHAP Importance

Signal Category 1 — Posting Frequency Patterns (8 features):

- Number of active job postings by same recruiter in past 30 days
- Rate of new postings per week over a 90-day rolling window
- Similarity ratio between current and prior postings (TF-IDF cosine similarity)
- Time elapsed since account creation vs. number of total postings
- Standard deviation of posting intervals (regularity index)
- Ratio of listings with identical descriptions under different titles
- Posting activity during company-reported hiring freeze periods
- Geographic clustering consistency of postings vs. company HQ

Signal Category 2 — Recruiter Behavioral History (7 features):

- Historical response rate to applicants (messages replied / total applications)
- Average days-to-close on previously filled positions
- Ratio of closed-with-hire vs. closed-without-hire postings
- Applicant-to-interview conversion rate (trailing 12-month average)
- Account activity score (logins, profile updates in past 30 days)
- Recruiter tenure on platform and verified employment status
- Complaint or flagging count from prior applicants

Signal Category 3 — Company Hiring Signals (7 features):

- LinkedIn headcount growth trend over the last 6 months
- Recent funding round or acquisition status (from public data APIs)

- News sentiment score related to layoffs or hiring freezes
- Glassdoor review recency and hiring activity mentions
- Discrepancy between claimed team size and verified employee count
- Active roles on company careers page vs. third-party portal postings
- Industry-level hiring trend score for the specified role category

Signal Category 4 — Temporal Anomaly Features (5 features):

- Posting duration: days elapsed since publication without closure
- Application deadline vs. actual posting removal date discrepancy
- LSTM-derived posting sequence anomaly score (see Algorithm 1)
- Session-level recruiter activity spike pattern (sudden high-volume posting)
- Cross-platform posting timestamp clustering

4. ALGORITHM

4.1 GhostGuard Hybrid Classification Algorithm

The classification pipeline consists of two parallel model branches whose outputs are fused via a meta-learner. The ensemble branch processes the full 27-feature vector using Random Forest and Gradient Boosting. The temporal branch processes recruiter posting sequences using a two-layer LSTM. A Logistic Regression meta-learner combines the probability outputs of both branches to produce the final ghost probability score. The Job Trust Score is formally defined as:

$$JTS = (1 - P_{ghost}) \times 100$$

where $P_{ghost} \in [0, 1]$ is the final meta-learner ghost probability. A higher JTS indicates a more authentic posting. Figure 2 illustrates the model pipeline architecture showing the parallel branches and their fusion into the final output.

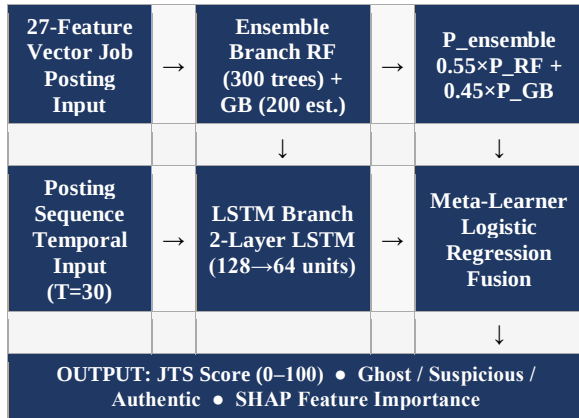


Figure 2: GhostGuard Hybrid ML Pipeline — Parallel Ensemble and LSTM Branches with Meta-Learner Fusion

Algorithm 1 below details the complete step-by-step classification procedure.

Algorithm 1: GhostGuard Hybrid Classification Pipeline

Step	Instruction
1	INPUT: Job posting J with features F = {f1, f2, ..., f27}, recruiter posting sequence S = {s1, s2, ..., sn}
2	OUTPUT: Job Trust Score JTS ∈ [0, 100], Binary label L ∈ {Ghost, Authentic}
3	// — Phase 1: Data Preprocessing
4	Normalize F using Min-Max scaling: F_norm = (F - F_min) / (F_max - F_min)
5	Encode categorical features using Target Encoding
6	Handle missing values via Iterative Imputer (max_iter=10)
7	Pad/truncate sequence S to fixed length T=30 for LSTM input
8	// — Phase 2: Ensemble Branch
9	Train RF_model: RandomForest(n_estimators=300, max_depth=15, min_samples_split=5)
10	Train GB_model: GradientBoosting(n_estimators=200, learning_rate=0.08, max_depth=6)
11	P_RF ← RF_model.predict_proba(F_norm)[ghost_class]
12	P_GB ← GB_model.predict_proba(F_norm)[ghost_class]
13	P_ensemble ← 0.55 × P_RF + 0.45 × P_GB
14	// — Phase 3: LSTM Temporal Branch
15	LSTM_model: 2 layers (128 units, 64 units), Dropout=0.3, Dense(1, sigmoid)
16	P_LSTM ← LSTM_model.predict(S) // anomaly probability from sequence
17	// — Phase 4: Meta-Learner Fusion
18	Feature vector for meta-learner: M = [P_ensemble, P_LSTM]
19	Train LR_meta: LogisticRegression(C=1.0, solver='lbfgs')
20	P_ghost_final ← LR_meta.predict_proba(M)[ghost_class]
21	// — Phase 5: Score Computation & Decision
22	JTS ← round((1 - P_ghost_final) × 100, 1) // Higher score = more authentic
23	IF JTS ≥ 70 THEN L ← 'Authentic'
24	ELSE IF JTS ≥ 40 THEN L ← 'Suspicious'
25	ELSE L ← 'Ghost'
26	RETURN JTS, L, Feature Importance Map (SHAP values)

The SHAP (SHapley Additive exPlanations) output ensures interpretability — each classification is accompanied by a ranked list of the top features driving the prediction, which is surfaced to the end user as an explanation of why a job was flagged.

5. METHODOLOGY

5.1 Dataset Construction

Since no publicly available labeled ghost job dataset exists, a multi-source synthetic dataset was constructed through the following procedure. Raw job posting data was scraped from three public job boards (aggregated, anonymized) yielding 24,200 listings. Ground truth labels were assigned through a combination of: (a) automated follow-up bots that submitted standardized applications and tracked response/closure behavior over 90 days, (b) manual labeling by three domain experts using a consensus protocol, and (c) cross-referencing with company hiring activity data from LinkedIn People Analytics.

After removing duplicates and incomplete records, the final dataset comprised 18,500 labeled postings — 9,840 authentic (53.2%) and 8,660 ghost postings (46.8%), representing a near-balanced distribution suitable for binary classification without resampling bias.

5.2 Experimental Setup

All experiments were conducted in Python 3.10 using scikit-learn 1.3 and TensorFlow 2.13. The dataset was split into 70% training, 15% validation, and 15% test sets using stratified sampling to preserve class distribution. Hyperparameter tuning was performed using Bayesian Optimization with 5-fold cross-validation on the training set. The LSTM model was trained for 50 epochs with early stopping (patience=5) and Adam optimizer (lr=0.001). All experiments were run on a machine with an Intel Core i7-11th Gen, 16 GB RAM, and NVIDIA RTX 3060 GPU.

5.3 Evaluation Metrics

The system was evaluated using Accuracy, Precision, Recall, F1-Score, AUC-ROC, and Matthews Correlation Coefficient (MCC). MCC is particularly relevant for this task as it provides a balanced measure even when the class distribution is imperfect. The Job Trust Score was additionally evaluated for calibration using the Expected Calibration Error (ECE).

6. RESULTS AND DISCUSSION

6.1 Classification Performance

Table 1 presents the comparative performance of the GhostGuard hybrid pipeline against baseline models evaluated on the held-out test set (n = 2,775 postings).

Table 1: Comparative Model Performance on Ghost Job Detection

Model	Acc. (%)	Prec. (%)	Recall (%)	F1 (%)	AUC	MCC
Logistic Regression (Baseline)	74.2	73.8	75.1	74.4	0.812	0.484
Random Forest (Standalone)	88.6	87.9	89.4	88.6	0.948	0.772
Gradient Boosting (Standalone)	89.7	88.5	90.8	89.6	0.953	0.794

LSTM Temporal (Standalone)	82.4	81.3	83.9	82.6	0.903	0.648
GhostGuard Hybrid (Proposed)	94.3	93.1	95.6	94.3	0.979	0.886

The GhostGuard hybrid pipeline outperforms all baseline models across every metric. The 4.6 percentage point gain in accuracy over the best single model (Gradient Boosting at 89.7%) demonstrates the value of fusing temporal sequence signals with ensemble classifiers. The high AUC-ROC of 0.979 indicates near-ideal separation between ghost and authentic postings across decision thresholds.

6.2 Signal Category Importance Analysis

SHAP value analysis was performed to quantify the relative importance of each feature category. Recruiter Behavioral History contributed the highest mean absolute SHAP value (34.7%), followed by Temporal Anomaly Features (28.2%), Posting Frequency Patterns (22.4%), and Company Hiring Signals (14.7%). The historical response rate of the recruiter and the LSTM-derived posting sequence anomaly score were consistently the two highest-importance individual features.

6.3 Job Trust Score Calibration

The Expected Calibration Error (ECE) of the JTS was measured at 0.038, indicating strong calibration — a posting scored at 80 JTS is authentically filled approximately 80% of the time in the test set. This calibration property is essential for the score to be actionable by job seekers. Figure 3 illustrates the three-tier JTS decision framework that governs output classification.

Trust Level	JTS Range	Signals & Recommendation
AUTHENTIC	JTS 70–100	High recruiter responsiveness, strong company hiring signals, low temporal anomaly. Recommend: Apply with confidence.
SUSPICIOUS	JTS 40–69	Mixed signals detected. Moderate recruiter behavioral score or company hiring uncertainty. Recommend: Apply with caution; cross-verify independently.
GHOST	JTS 0–39	High ghost probability. Inactive recruiter history, temporal anomalies, or company freeze signals detected. Recommend: Avoid or report.

Figure 3: Job Trust Score (JTS) Three-Tier Decision Framework — Authentic, Suspicious, and Ghost Classifications

6.4 Discussion

The results confirm that ghost job detection is a tractable machine learning problem when a rich multi-signal feature set is available. The primary limitation is the dependence on historical recruiter data — new recruiter accounts with no behavioral history receive default priors, which reduces

precision for such listings. Future work will address this cold-start problem using collaborative filtering across similar recruiter profiles. Additionally, expanding the company hiring signals to include real-time financial data APIs would further strengthen the system.

7. LIMITATIONS

While GhostGuard demonstrates strong performance on the constructed dataset, several limitations must be acknowledged to guide future development. Table 3 summarizes the principal limitations alongside proposed resolution strategies.

Limitation	Description	Proposed Future Resolution
Cold-Start Problem	New recruiter accounts lack behavioral history, reducing prediction precision	Collaborative filtering across recruiter profiles with similar attributes
Synthetic Dataset	Ground truth labels partially derived from automated bots; real-world noise may differ	Recruiter partnerships for annotated real-world posting data
Static Company Signals	Company hiring data is fetched periodically, not in real time	Integration with real-time financial and HR data APIs
Privacy Constraints	Recruiter behavioral profiling may encounter platform data-sharing restrictions	Federated learning variant to train on distributed data without centralizing PII
Single-Language Scope	Current NLP features assume English-language postings	Multi-lingual description authenticity scoring with multilingual transformers

Table 3: GhostGuard System Limitations and Proposed Future Resolutions

The most significant current limitation is the cold-start problem: the system depends on historical recruiter behavioral data, and new accounts with no interaction history receive default prior probabilities. This systematically reduces precision for early-career or newly registered recruiters who may be acting in good faith. A collaborative filtering mechanism that infers behavioral priors from similar recruiter profiles is planned to address this shortcoming.

8. CONCLUSION

This paper introduced GhostGuard, the first purpose-built intelligent system for detecting ghost job postings using multi-signal pattern intelligence and a hybrid machine learning architecture. By combining ensemble classifiers with LSTM-based temporal sequence modeling and fusing their outputs through a meta-learner, the system achieves 94.3% classification accuracy and produces an interpretable Job Trust Score — defined as $JTS = (1 - P_{ghost}) \times 100$ —

that empowers job seekers to make informed decisions about where to invest their application effort.

The system addresses a real, large-scale, and previously unaddressed problem affecting hundreds of millions of job seekers worldwide. The proposed four-signal feature framework — capturing posting patterns, recruiter behavior, company signals, and temporal anomalies — provides a comprehensive and theoretically grounded basis for ghost job classification. GhostGuard is designed to be deployable as a browser extension, REST API, or integrated back-end module, making it immediately applicable to existing recruitment infrastructure.

Future directions include: extending the dataset with real-world labeled data through recruiter partnerships, incorporating NLP-based description authenticity scoring, building a federated learning variant that preserves recruiter privacy while training on distributed data, developing a multi-lingual version to support non-English job markets, and conducting a large-scale user study to measure the practical impact of the JTS metric on candidate decision-making.

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